INCH-POUND
MIL-DTL-81381/12D
17 February 2017
SUPERSEDING
MIL-W-81381/12C
5 October 1977

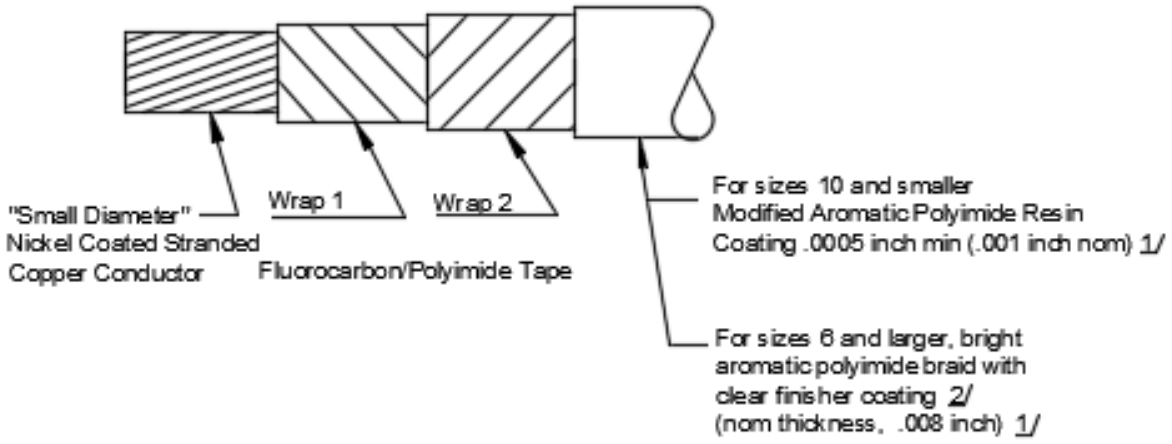
DETAIL SPECIFICATION SHEET

WIRE, ELECTRIC, FLUOROCARBON/POLYIMIDE INSULATED, MEDIUM WEIGHT, NICKEL COATED COPPER CONDUCTOR, 600 VOLTS, 200 DEGREES C, NOMINAL 8.4 OR 15.4 MIL WALL

Inactive for new design after 16 June 1997. For new design, use SAE AS22759/187 and AS22759/184 (for AWG size 8 and larger).

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-81381.



NOTES:
1/ Nominal values are for information only. Nominal values are not requirements.
2/ Braid (Sizes 8 and larger): Bright aromatic polyamide yam, 200 denier, 100 filaments, tightly formed, uniform in appearance, treated with a clear finisher coating. The finisher coating shall be compatible with the 200° C temperature rating and the performance requirements of the insulated wire.

FIGURE 1 - Wire construction.

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TABLE I. Construction details.

Part or Identification Number (PIN) 1/	Wire Size	Conductor			Finished Wire				Insulation Tapes			
		Stranding (Number of Strands x AWG Gage of Strands	Diameter (inches)		Resistance at 20° C (68° F) (ohms/1000 ft.) (max)	Diameter (inches) (min – max)	Weight (lbs/1000 ft.)		Wrap 1		Wrap 2	
			min	max			nom. 2/	max	Tape Code 3/	Overlap (%) (min)	Tape Code 3/	Overlap (%) (min)
M81381/12-24-*	24	19 x 36	.023	.024	25.9	.040 - .045	2.1	2.2	0/2/.5	50	.1/1/.1	50
M81381/12-22-*	22	19 x 34	.029	.030	16.0	.045 - .050	3.0	3.2				
M81381/12-20-*	20	19 x 32	.037	.038	9.77	.053 - .058	4.6	4.8				
M81381/12-18-*	18	19 x 30	.046	.048	6.10	.063 - .068	6.8	7.2				
M81381/12-16-*	16	19 x 29	.052	.054	4.76	.068 - .074	8.6	9.0				
M81381/12-14-*	14	19 x 27	.065	.068	3.00	.081 - .087	13.1	13.8				
M81381/12-12-*	12	37 x 28	.084	.087	1.98	.100 - .107	19.9	20.9				
M81381/12-10-*	10	37 x 26	.106	.110	1.24	.122 - .129	30.7	32.4				
M81381/12-8-*	8	133 x 29	.158	.166	.694	.188 - .206	58.7	61.8			.5/1/.5	50
M81381/12-6-*	6	133 x 27	.198	.208	.436	.228 - .251	90.4	95.1				
M81381/12-4-*	4	133 x 25	.250	.268	.275	.280 - .306	141	148				
M81381/12-2-*	2	655 x 30	.320	.340	.177	.350 - .378	224	235				

1/ PIN: The asterisks in the PIN column of Tables I and II shall be replaced by color code designators in accordance with MIL-STD-681, except that opaque dark yellow as defined in MIL-DTL-81381 shall be designated by the letter “N” and unpigmented polyimide resin coating shall be designated by the letter “C”. Examples: Size 20, opaque dark yellow – M81381/12-20-N, same with orange stripe – M81381/12-20-N3. A non-white insulation may be used with striping or banding (certain color combinations with low color contrast can make identification difficult or increase the chance of misidentification).

2/ Nominal values are for information only. Nominal values are not requirements.

3/ Tape Codes:

0/2/.5 2 mil polyimide film/0.5 mil FEP fluorocarbon resin

.1/1/.1 0.1mil FEP fluorocarbon resin/1 mil polyimide film/0.1 mil FEP fluorocarbon resin

.5/1/.5 0.5mil FEP fluorocarbon resin/1 mil polyimide film/0.5 mil FEP fluorocarbon resin

FEP = Fluorinated Ethylene Propylene

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TABLE II. Performance details.

Part or Identification Number (PIN)	Durability Test Load for		Abrasion Resistance				Bend Testing					
							Mandrel diameter (inches) (±3%)				Test Load (lbs) (±3%)	
	Color Marking (grams)	Insulation Coating (lbs)	Weight Support Bracket	Weight (lbs)	Tension Load (lbs)	Resistance (inches of tape) (min)	Life Cycle (Oven & bend tests) 2/	Cold Bend Test	Wrap Test	Wrinkle Test	Life Cycle (Oven & bend tests) 2/	Cold Bend Test
M81381/12-24-*	75	.75	A	1.0	1.0	12	.250	.250	.125	.156	.500	
M81381/12-22-*	100	1.00	A	1.0	1.0	14	.250	.250	.125	.188	.750	
M81381/12-20-*	100	1.00	A	1.0	1.0	16	.250	.250	.125	.250	.750	
M81381/12-18-*	150	1.00	A	1.0	1.0	16	.375	.375	.250	.312	1.00	
M81381/12-16-*	150	1.00	A	1.0	1.0	20	.375	.375	.250	.375	1.00	
M81381/12-14-*	150	1.00	B	1.0	2.0	24	.500	.375	.375	.500	2.00	
M81381/12-12-*	150	1.00	B	1.0	2.0	26	.750	.750	.375	.750	2.00	
M81381/12-10-*	150	1.00	B	1.0	2.0	26	.750	.750	.375	1.00	3.00	
M81381/12-8-*	150	1/	C	4.25	2.0	18	2.00	2.00	1.00	1/	3.00	
M81381/12-6-*	150	1/	C	4.25	2.0	27	4.50	4.50	2.00	1/	3.00	
M81381/12-4-*	150	1/	C	4.25	2.0	27	6.00	6.00	3.00	1/	4.00	
M81381/12-2-*	150	1/	C	4.25	2.0	27	6.00	6.00	4.00	1/	6.00	

1/ Test not applicable
2/ Also for bend tests after immersion.

WIRE RATINGS AND ADDITIONAL REQUIREMENTS

Temperature rating: 200º C (392º F) max conductor temperature
Voltage rating: 600 volts (RMS) at sea level
Blocking: Oven temperature, 200 ±2º C (392 ±3.6º F)
Braid Fraying: The aromatic polyamide braid with finisher (wire sizes 8 and larger) shall be non-fraying and shall be firmly adherent to the underlying insulation. This examination shall be included in the MIL-DTL-81381 quality conformance inspection as a Group II characteristic, one specimen to be examined from each sample unit.
Color — Sizes 24 through 10: As specified in contract or order in accordance with MIL-DTL 81381.
Sizes 8 through 2: The natural color of the finisher-treated aromatic polymide braid (off-white to amber) is preferred (color designator “C”). Other colors, varying with the supplier, are available. The MIL-STD-104 color limits and the extended color limits of MIL-DTL 81381 are not applicable to these sizes.
Flammability: 3 sec (max) after flame
3.0 inches (max) flame travel
No flaming of tissue paper
Humidity Resistance: 5 megohms-1000 ft, min insulation resistance after humidity exposure
Identification of product: Required for sizes 22 AWG and larger
Identification, striping, or banding durability: 125 cycles (250 strokes); see Table II for test load
Impulse dielectric test: 100% test; impulse voltage as specified in MIL-DTL-81381
Insulation resistance: 2500 megohm-1000 ft. (min)
Lamination sealing: Oven temperature 230±2º C (446±3.6º F)
Life cycle: Oven temperature 230±2º C (446±3.6º F) for 500 hours
Minimum wall thickness: Sizes 24 through 10: 8.0 mils
Sizes 8 through 2: 15.0 mils
Polyimide cure test: Applicable to sizes 10 and smaller
Propellant resistance: Test not required

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Resin coating durability: Sizes 24 through 10: 250 cycles (500 strokes); see Table II for test load
Sizes 8 through 2: Test not applicable
Shrinkage: 0.031 inches (max) at 230±2º C (446±3.6º F)
Surface resistance: 5 megohms-inches (min), initial and final readings
Thermal shock: Oven temperature, 200 ±2º C (392 ±3.6º F), Change in measurement: 0.031 inches (max)
Wet dielectric test: 2500 volts (RMS)
Wrinkle test: Applicable to sizes 10 and smaller. No wrinkles shall be visible in the insulation at 3X magnification (3 diameters) after bending the wire one full turn around the mandrel specified in Table II (The wire may be examined on the mandrel or after removal of the mandrel leaving the coal intact.) This test shall be included in the MIL-DTL-81381 quality conformance inspection as Group II characteristic, one specimen to be tested from each sample unit.

Caution: This wire should not be subjected to physical contact with missile propellants.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced document. In addition to the basic specification MIL-DTL-81381, this document references:
MIL-STD-681
AS22759/184
AS22759/187

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

Review activities:
Army – AR, AT, MI
Navy – EC, MC, OS, SH

(Project 6145-2016-053)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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